Unit I

An automation is defined as a system where energy materials & information are transformed, transmitted & used for performing some functions without human intervention.

The abstract machines according to their increasing capabilities like

① Basic machine
② Finite state machine
③ Turing Machine

Basic machine :- It is the machine with a set of i/ps I & set of output O. It maps the i/p set I to the o/p set O. i.e for specific i/p I, it produces specific o/p O. This mapping function is called as MAchine Function (MAF)

$$I \rightarrow O \quad (MAF)$$

e.g. consider a vending m/c which gives chocolate as an o/p. The following table shows m/c function MAF for the same.

I	O
Rs. 5.00	Five star
Rs. 10.00	Milky bar
Rs. 20.00	Bar one
Rs. 50.00	Dairy Milk

* Introduction to Finite Automata (FA) :

This is the most general model with the internal or intermediate states of the machine

e.g. Game of snake & ladder.

Defination of FA :-

It consists of finite set of states & a set of transistions from one state to another state on i/p symbols selected from an alphabet Σ.

Initial state is generally denoted by q_0.

FA is defined by a 5 tuple $(Q, \Sigma, \delta, q_0, F)$

Where,

Q - Finite set of states.

Σ - Finite set of i/p symbols

δ - Transition function mapping $Q \times \Sigma$ to Q

q_0 - Initial state & $q_0 \in Q$

F - Set of final states.

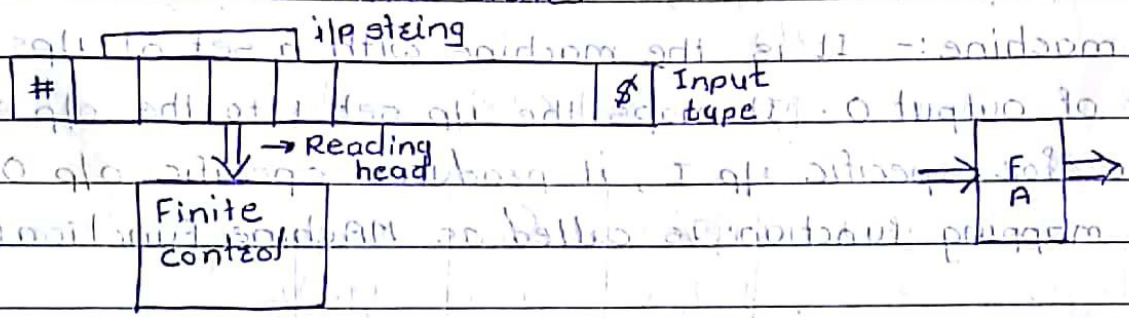


fig: Block diagram of FA

The FA has i/p tape divided into units each containing one symbol of i/p string which is built from symbols of alphabet Σ.

The start of tape is indicated by # & end is indicated by $. Tape may be infinite or finite.

In case of infinite the tape end marker is absent

The i/p string is to be processed is stored from left to right direction.

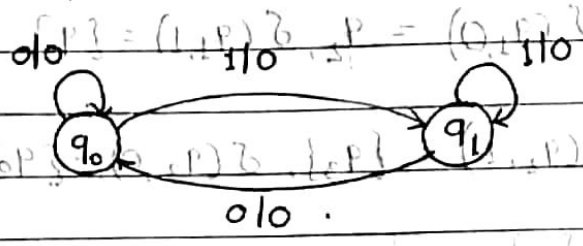
The reading head examines one unit at a time & can move one unit in only one direction generally left to right.

Representation:-

FA can be represented using graph or funct^n notation using adjacency matrix.

Transition graph:-

Transition diagram is finite directed lebelled graph in which vertices represent a state & directed edges indicate the transition of a state. & edges are lebelled with i/p or o/p.



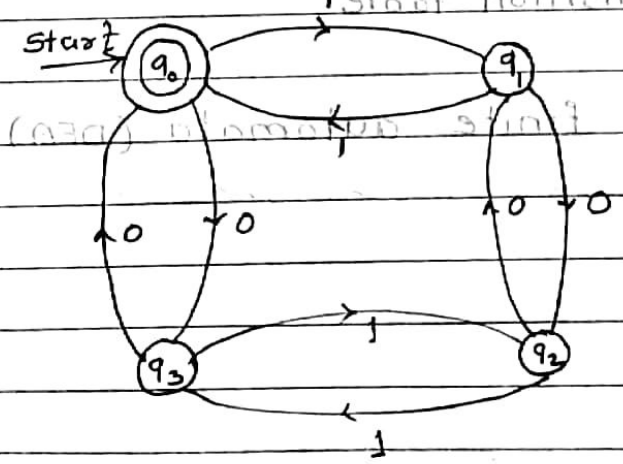
Finite state m/c consists of a pair of functions namely:-

$I \times S \rightarrow O$ (MAF) (MAchine Function) &

$I \times S \rightarrow S$ (STF) (STate Function).

MAF maps the cartetion product of i/p set I & set of states S to the set of o/p O.

STF maps cartetion product of i/p set I & state set S to the state set S.



Final state is shown by double concentric circles.

single concentric circle is for non-final state.

q_0 is initial state, therefore pointed by an arrow along with the string "start".

q_0 is also final state therefore, shown by double circle.

$$M = (\{q_0, q_1, q_2, q_3\}, \{0,1\}, \delta, q_0, \{q_0\})$$

$\delta(q_0, 0) = \{q_3\}$

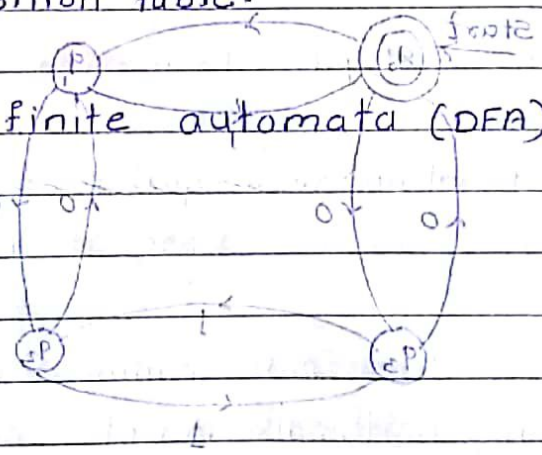
$\delta(q_0, 1) = \{q_1\}$, $\delta(q_1, 0) = q_2$, $\delta(q_1, 1) = \{q_0\}$

$\delta(q_2, 0) = \{q_1\}$, $\delta(q_2, 1) = \{q_3\}$, $\delta(q_3, 0) = \{q_0\}$

$\delta(q_3, 1) = \{q_2\}$

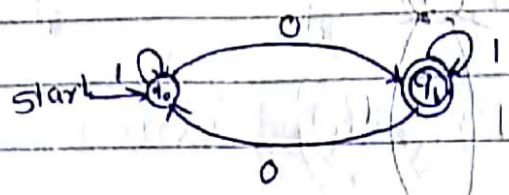
	0	1
q_0	q_3	q_1
q_1	q_2	q_0
q_2	q_1	q_3
q_3	q_0	q_2

Transition table:

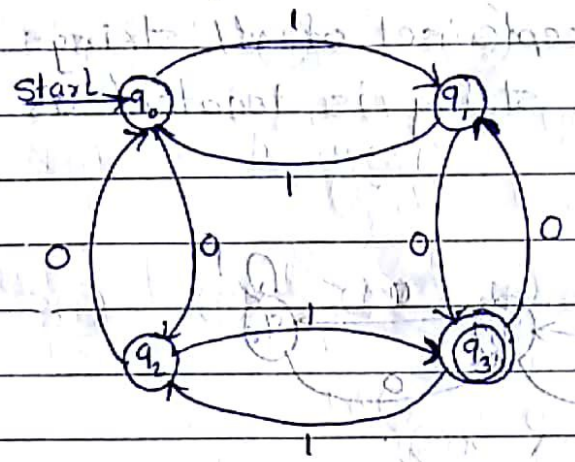


DFA :- Deterministic finite automata (DFA).

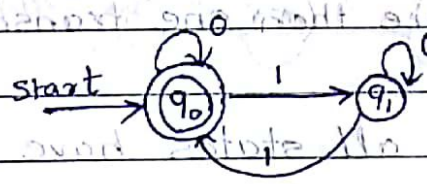
Design DFA which accepts all strings containing odd no. of 0's



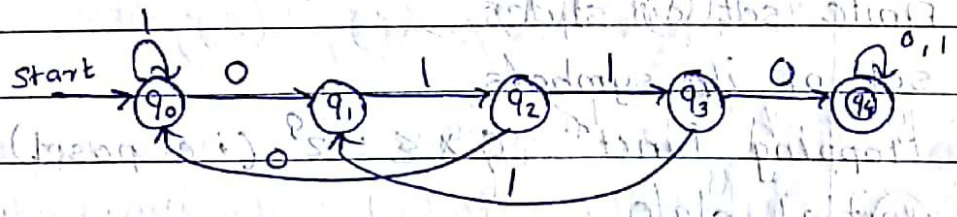
Design DFA which accepts set of all strings containing odd no. of 0's & odd no. of 1's



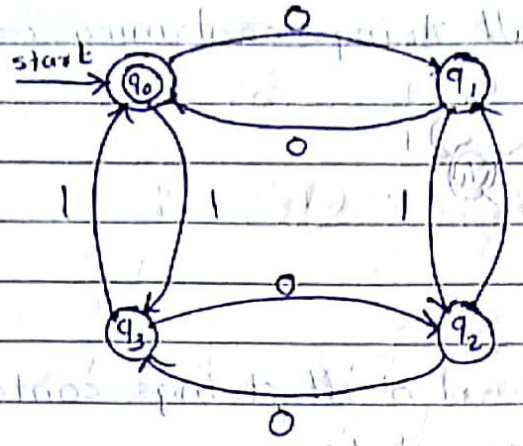
Design DFA which accepts set of all strings containing even no. of 1's



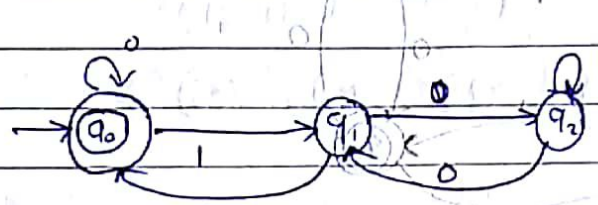
Design DFA which accepts all strings contains 10110 as substring where $\Sigma = \{0, 1\}$



Design DFA which accepts set of strings containing even no. of 0's & 1's



Design DFA which accepts set of all strings containing mod3 is zero where string is treated as binary



NFA :- Non Deterministic Finite Automata

① one i/p symbol causes to move more than one state or a state may contain more than one transition from same i/p symbol.

② It is not necessary that all states have to consume all i/p symbols in Σ.

NFA can be defined by 5 tuples

$M = (Q, \Sigma, \delta, q_0, F)$

Q = Finite set of states.

Σ = set of i/p symbols

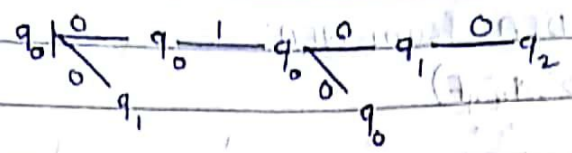
δ = Mapping funct^n $Q \times \Sigma \to 2^Q$ (i.e. power set)

q_0 = initial state

F = Finite set of final states.

Acceptance of NFA :-

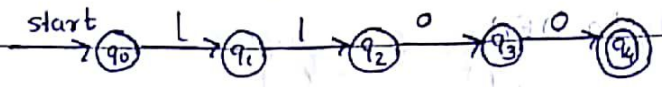
consider a string 0100. Let us check the acceptability



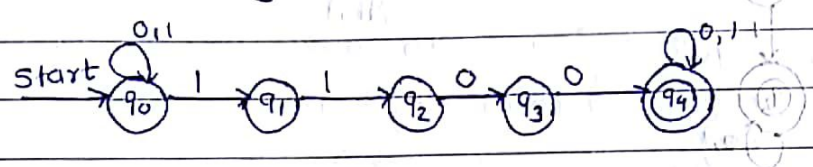
q_2 is a final state. Hence string is accepted.

① Design an NFA which accepts set of strings over alphabet set {0, 1} & ending with 2 concecutive 0's.

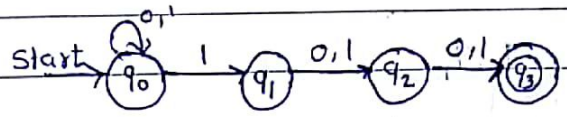
② Design NFA which accepts string 1100 only



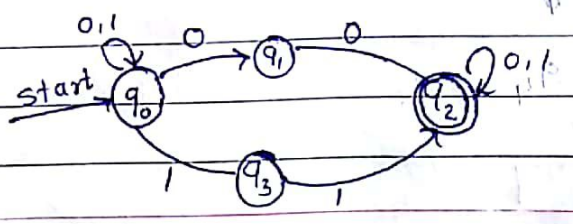
③ Design NFA which accepts set of all strings containing 1100 as substring



④ Design NFA which accepts set of all strings containing 3rd symbol from right side is 1.



⑤ Design NFA to accept the strings with 0's & 1's such that string contains either two concecutive 0's or two concecutive 1's.



Algorithm to convert DFA from NFA:-

Let NFA is $M = (Q, \Sigma, \delta, q_0, F)$

equivalent DFA is

$M' = (Q', \Sigma', \delta', q_0', F')$

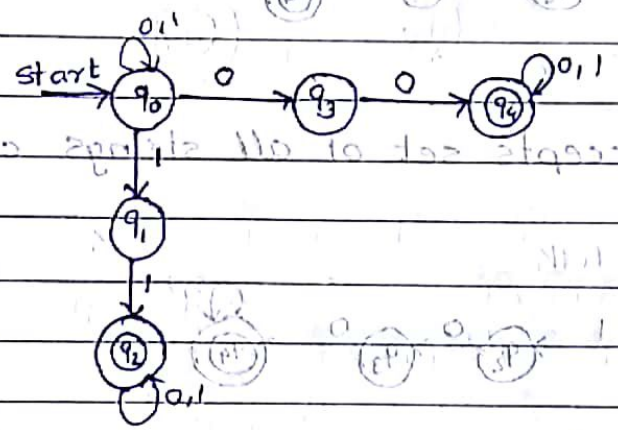
Q is calculated as 2^Q

$F' \subseteq Q$, which contains set of final states for given NFA

δ' can be calculated as,

$$\delta'(q, a) = \delta(q, a) \text{ \& } \delta([q_0, q_1], a)$$
$$= \delta(q_1, a) \cup \delta(q_0, a)$$

e.g. Convert NFA to DFA.



Transition table:-

state \ I/P s	0	1
q_0	q_0, q_3	q_0, q_1
q_1	ϕ	q_2
q_2	q_2	q_2
q_3	q_4	ϕ
q_4	q_4	q_4

First we make starting state of NFA as starting state of DFA.

DFA's transition table:-

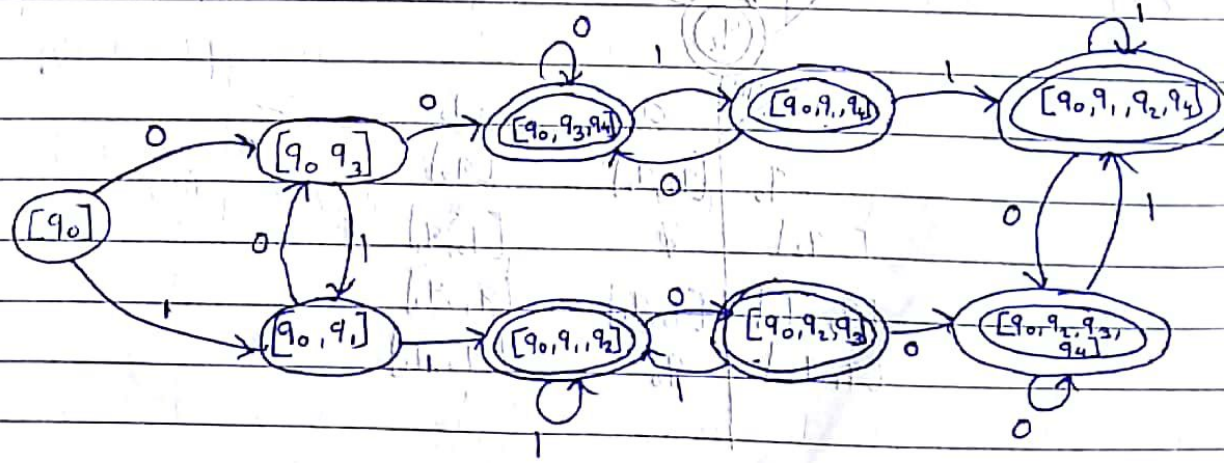
For DFA M_2 equivalent to above M_1 defined as,

$$M_2 = (2^Q, \{0,1\}, \delta, [q_0], F)$$

$$F = \{[q_0, q_3, q_4], [q_0, q_1, q_4], [q_0, q_1, q_2, q_4], [q_0, q_1, q_2], [q_0, q_2, q_3], [q_0, q_2, q_3, q_4]\}$$

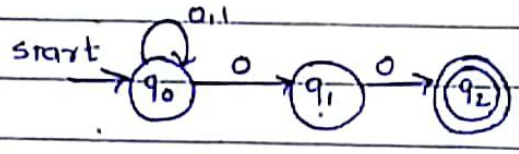
δ is defined as

state \ i/p	0	1
$[q_0]$	$[q_0, q_3]$	$[q_0, q_1]$
$[q_0, q_2]$	$[q_0, q_3, q_4]$	$[q_0, q_1]$
$[q_0, q_1]$	$[q_0, q_3]$	$[q_0, q_1, q_2]$
$[q_0, q_3, q_4]$	$[q_0, q_3, q_4]$	$[q_0, q_1, q_4]$
$[q_0, q_1, q_2]$	$[q_0, q_2, q_3]$	$[q_0, q_1, q_2]$
$[q_0, q_1, q_4]$	$[q_0, q_3, q_4]$	$[q_0, q_1, q_2, q_4]$
$[q_0, q_2, q_3]$	$[q_0, q_2, q_3, q_4]$	$[q_0, q_1, q_2]$
$[q_0, q_1, q_2, q_4]$	$[q_0, q_2, q_3, q_4]$	$[q_0, q_1, q_2, q_4]$
$[q_0, q_2, q_3, q_4]$	$[q_0, q_2, q_3, q_4]$	$[q_0, q_1, q_2, q_4]$

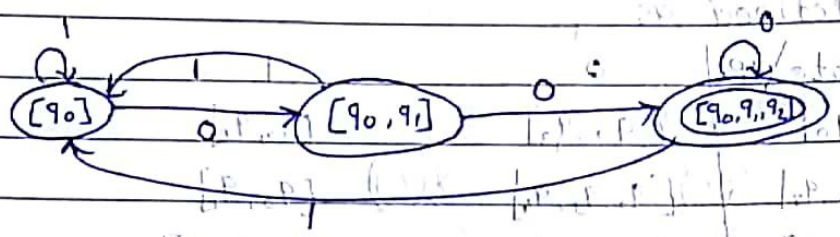


DFA.

Convert NFA to DFA:-

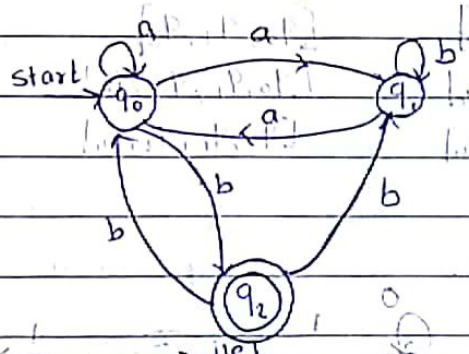


	0	1
$[q_0]$	$[q_0, q_1]$	$[q_0]$
$[q_0, q_1]$	$[q_0, q_1, q_2]$	$[q_0]$
$[q_0, q_1, q_2]$	$[q_0, q_1, q_2]$	$[q_0]$

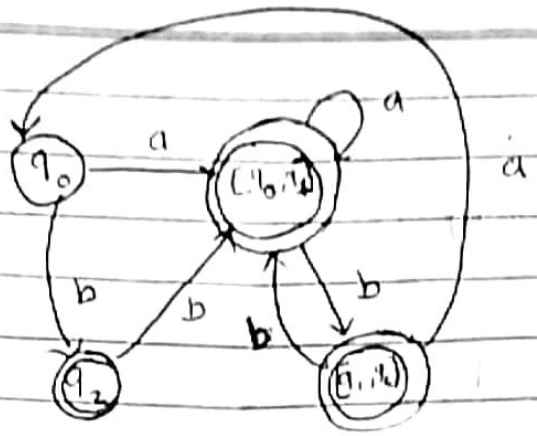


DFA.

Construct a DFA diagram to NFA



States	a	b
q_0	$[q_0, q_1]$	$[q_2]$
$[q_2]$	ϕ	$[q_0, q_1]$
$[q_0, q_1]$	$[q_0, q_1]$	$[q_1, q_2]$
$[q_1, q_2]$	q_0	$[q_0, q_1]$


a
a
a
b
b
b
b